COL1000: Lab 5

Semester 2, 2025–26

Instructions Upload your code to Gradescope as: <entryNumber>_q<quesNumber>.py. After uploading, run the autograder to check that all tests pass. Make sure not to include any custom prompt text in `input()` calls, and any extra text in `print()` statements. You must use only basic control structures (loops, recursion, conditionals, functions) and avoid advanced built-in python functions.

Question 1: Sum of factors Write a code that reads a positive integer `n` and prints sum of all proper factors of `n` (all factors except `n` itself) using a function `sum_factors(n)`. For example, for `n=12`, proper factors are `1,2,3,4,6`, so output must be `16`.

Question 2: Factorial computation Write a code that reads a positive integer `n` and prints its factorial using a recursive function `factorial(n)`. For example, for the input `n=5` the code should print `120`. Your code must not use `while` and `for` loops.

Question 3: Binomial coefficient Write a code that reads two positive integers `n, k` and prints the binomial coefficient $\binom{n}{k}$ using a recursive function `binomial(n, k)`. Your code must not use `while` and `for` loops.

Hint: Use the recursive relation:

$$\binom{n}{k} = \begin{cases} 1 & \text{if } k = 0 \text{ or } k = n \\ \binom{n-1}{k-1} + \binom{n-1}{k} & \text{if } 0 < k < n \end{cases}$$

Question 4: Subsequence check Write a code that reads two non-empty strings `A`, `B`, and uses recursive function `is_subsequence(A, B)` to check if `B` is a subsequence of `A`. The code should print `True` if `B` is found, and `False` otherwise. For example, `union` is a subsequence of `function`.

Hint: To check if `A` is a subsequence of `B` compare the first characters of both. If they match, recurse on the remaining parts of both strings `(A[1:], B[1:])`. If they don't match, skip the first character of `A`.